Total-load observability of continuous girder bridges from a single deflection channel: closed-form blind configurations, laboratory verification and sensor placement on an in-service box girder

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Abstract

Influence-area bridge weigh-in-motion estimates gross vehicle weight from the area under one response record. On continuous girders the influence line changes sign, and a sensor may record almost no net area. This paper treats this as a sensor-placement question for structural health monitoring: where can one deflection channel observe the total load? For a linear-elastic beam the deflection integral over a complete crossing equals the total load times a constant C_m, which by reciprocity is the sensor deflection under a uniform unit load. This gives closed forms for two- and three-span beams. With uniform stiffness, a mid-span sensor on two spans is blind at the span ratio $(3+\sqrt{33})/6 = 1.45743$; a mid side-span sensor on a symmetric three-span beam is blind at $\beta = 1.700594$, and the side-span blind locus ends at $\beta = 2\cos(\pi/9) = 1.879385$. Laboratory tests on a two-span steel beam identified two- and three-load groups within 1.19 % (beam-property method) and 0.70 % (standard-load method). On a 55 + 90 + 55 m box girder, two calibrations reproduce the acceptance-test deflections but give a side-span gauge index of 0.18 or 0.48, while the centre-span gauge stays at 0.21–0.23. A residual offset of 0.05 mm then biases a 25.06 t truck by 35–93 % at the side-span gauge and 10–11 % at the centre-span gauge. Side-span gauges on continuous girders therefore need C_m from an in-situ identified model or a weighed-vehicle calibration, not from the design model.

Keywords: Structural health monitoring; Sensor placement; Bridge weigh-in-motion; Continuous girder; Deflection integral; Load observability

Nomenclature

Symbol	Meaning
$A^{+¿¿}$, A^{-}	positive and negative parts of the area under η_m (mm²/N)
C_m	invariance constant at sensor m, the signed area under η_m (mm²/N; in Section 4 m²/kN, 1 m²/kN = 10³ mm²/N)
EI, EI_{ref}	flexural rigidity; reference rigidity used for normalisation (N mm²)
I_y, I_t	deflection integral over a complete crossing in space (mm²) and in time, $I_t = I_y / v$ (mm s)
k	rotational stiffness of a monolithic inner pier (kN m/rad)
L_1, L_2	instrumented and adjacent span of a two-span beam (mm)
L_s, L_m	side and main span of a symmetric three-span beam (m)
L_{tot}	length of the integration record (mm; m in Section 6.2 for the bridge)
N_m	normalised indicator $192\,EI_{ref} C_m / L^4$
P_{tot}, P_0, P_t	total load; reference load; unknown load (N)
P_ε	offset-equivalent load $\varepsilon_0 L_{tot} / C_m$ (N; kN in Section 6.2)
R_m	cancellation ratio $C_m / A^{+¿¿}$
T_{cr}, T_1	crossing time; fundamental period (s)
v	traverse speed (mm/s)
α	ratio of effective to design flexural rigidity, EI_{eff} / EI_{design}
β	span ratio L_m / L_s
γ	relative gain error of the deflection chain
ε_0	residual zero offset of the deflection signal after pre/post re-zeroing (mm)
$\eta_m(x)$	deflection influence line at m (mm/N)
μ, ν	interior-moment parameters, Eqs. (5), (8) and Appendix D
ξ	sensor position as a fraction of the instrumented span
ρ_L, ρ_{EI}	span ratio and stiffness ratio, adjacent span over instrumented span
BPM, SLM	beam-property method; standard-load method

1. Introduction

Heavy vehicles above the design envelope shorten the life of highway bridges and, in extreme cases, contribute to collapse. A span of the Yangmingtan Bridge failed in 2012 under four overloaded trucks crossing together [1]. In other well-documented failures, heavy traffic acted together with design deficiencies or material degradation [2–4]. Knowing the loads that a bridge actually carries is therefore part of risk-informed asset management [5,6].

Pavement weigh-in-motion (WIM) is a mature technology [7,8] with codified accuracy classes [9], but stations are costly to install and maintain [10], coverage is sparse, and a pavement station weighs the vehicle rather than loading any particular bridge. Bridge weigh-

in-motion (BWIM), introduced by Moses [11], instruments the structure and fits the measured response to the sum of axle loads times influence-line ordinates; it requires axle spacings and speed. The approach now includes influence lines calibrated from measurements [12], regularised solutions [13], systematic accuracy improvements [14,15], contactless variants [16] and virtual-axle methods that avoid axle detectors [17]. In parallel, moving-force identification recovers the axle-load history in the time or frequency domain [18–20], on multi-span bridges with elastic bearings [21], and with dynamic axle loads [22], steepest-descent [23], Bayesian [24], Kalman-filter [25] and physics-informed machine-learning [26] formulations. These methods aim at axle loads and need several sensors, a structural model or a regularised inversion.

A simpler family estimates only the gross weight from the area under the measured response. Ojio and Yamada, as described by Cardini and DeWolf [27], multiply the area under the strain–time record by the speed to obtain an “influence area” and calibrate it against a truck of known weight. On a seven-span slab-on-girder bridge in Winnipeg this influence-area method was accurate to within 5 % for the test trucks when their speeds were estimated accurately [28,29]; the review [29] also records a negative gross weight there, but from a different, asymmetry-based method, and neither source we could consult states whether the instrumented span was continuous. Related strain-area and strain-integral formulations have since been proposed [30–33], simplified BWIM without axle detectors has been verified in the field [34], and the same idea has been applied to support reactions measured by load cells [35]. The integral of the displacement influence line has also been used as a damage feature, where the ratio of two sensors removes the load [36], and a single-sensor strain feature grounded in influence-line theory has been applied to defect diagnosis, with a case study on a two-span continuous bridge [37]. The present authors earlier used the integral of the mid-span deflection to identify the total load on a simply supported beam [38], where the influence line has one sign and no configuration is blind. The relation between response area and gross weight is therefore established; we use it as the starting point.

What this literature does not address is the observability of the total load on a continuous girder. Over an interior support the influence line changes sign, so the positive area of the loaded span and the negative area of the adjacent span partly cancel. Distributed fibre-optic strain on a five-span continuous bridge has been used to obtain gross weight without influence lines [39], with many sensors along the bridge. Optimal sensor placement for BWIM has been addressed with an augmented Kalman filter formulation [40], and load identification from

multi-point bridge deflection has been formulated as a regularised stochastic problem [41]. Field WIM from a vision-based mid-span deflection, with end-mounted accelerometers for axle detection, has also been reported [42], using a calibrated influence line rather than the integral. Measured deflection influence lines from slow passes of a single heavy vehicle have recently been inverted for the sectional bending stiffness of variable-section continuous girders [43]. The question asked here is narrower and comes first: for a given continuous girder and sensor position, can a single deflection channel observe the total load at all? In monitoring terms it is a sensor-placement question that precedes the choice of any weighing algorithm.

The question matters because continuous girders are common. Two-span overpasses are a large population of highway structures, and three-span units with a long main span are the usual form of medium-span box girders built by balanced cantilever. On both, a sensor on one span responds negatively to load on the adjacent span, so the area-based weighing that works on simple spans can degrade, or fail, without any sign of it in the individual readings.

The contributions are as follows. First, the invariance of the deflection integral is stated for any linear-elastic beam, and the constant C_m is identified by reciprocity as the sensor deflection under a uniformly distributed unit load; the identity itself is classical, and we did not find it used in this way in the BWIM literature we reviewed. This result is the foundation, not the headline. Second, closed forms of C_m are derived for two-span beams with arbitrary span and stiffness ratios and for symmetric three-span beams, including the sign change of C_m . Third, the blind configurations, where a sensor records zero net integral for any vehicle, are located in closed form, together with the exact optimal sensor position. Fourth, $|C_m|$, the cancellation ratio $C_m / A^{+¿¿}$ and the offset-equivalent load P_ε are proposed as pre-installation observability indicators. Fifth, an error budget specific to the integral estimator shows that the residual zero offset, not random noise, dominates. The theory is verified on a laboratory two-span beam with statically positioned loads and applied to the acceptance-test data of an in-service 55 + 90 + 55 m box girder. That application asks how far the observability of each installed gauge can be determined from design data and two static readings. It is not a validation of the integral identity on a continuous bridge, because no crossing record exists for that structure.

103 “Observability” is used throughout in the narrow sense of total-load observability: a non-zero
104 C_m, so that the total load maps to a non-zero integral. It is not the observability of a state-
105 space model.

106 Section 2 develops the theory. Section 3 reports the laboratory verification, Section 4 the
107 application to the in-service bridge, and Section 5 a short analysis of field crossing records.
108 Section 6 gives the error budget and the dynamic range of validity, Section 7 the
109 recommendations and limitations, and Section 8 the conclusions.

110 2. Theory

111 This section states the invariance property, derives C_m for two- and three-span beams, and
112 defines the two estimators and the observability indicators; derivations are in Appendix A.

113 *2.1. Invariance of the deflection integral*

114 Consider a linear-elastic beam of total length L with time-invariant properties, and a fixed
115 measurement point m. Let $\eta_m(x)$ be the deflection at m due to a unit vertical force at abscissa
116 x, with $\eta_m = 0$ outside $[0, L)$. A convoy of n forces P_i at fixed offsets a_i from the leading force
117 ($a_1 = 0$) crosses the beam quasi-statically. By superposition the deflection at m when the
118 leading force is at x is

$$y_m(x) = \sum_{i=1}^{n} P_i \eta_m(x - a_i). \tag{1}$$

119 Integrating Eq. (1) term by term and substituting $u = x - a_i$ maps every term onto the same
120 interval, because η_m vanishes outside $[0, L)$:

$$I_y = \int_0^{L+a_{max}} y_m(x)\,dx = \left(\sum_{i=1}^{n} P_i\right) \int_0^{L} \eta_m(u)\,du = C_m P_{tot}. \tag{2}$$

121 The constant C_m depends only on the structure and the sensor position, not on the number of
122 axles, their loads or their spacings. The integration must run from the entry of the leading axle
123 to the exit of the trailing one, $[0, L + a_{max})$. Truncating at $x = L$ removes the tail of every
124 trailing axle, and on a continuous beam that tail lies where η_m is negative, so the error does
125 not cancel.

The constant has a second, more useful form. Let $w_m(x)$ be the deflected shape under a unit load at m, and $w_q(x)$ the deflection under a uniformly distributed load of intensity q_0 over the whole length. Maxwell's reciprocal theorem gives $\eta_m(x) = w_m(x)$. Betti's theorem applied to the unit point load at m and the uniform load gives $q_0 \int w_m \, dx = w_q(x_m)$. Both are standard results of structural analysis [44]. Hence

$$C_m = \int_0^L \eta_m \, du = \int_0^L w_m \, dx = \frac{w_q(x_m)}{q_0}. \tag{3}$$

The invariance constant is thus the deflection at the sensor under a uniform load over the whole structure, divided by the load intensity. It can be read from a standard table, obtained from one static analysis of an existing model, or measured with one static uniform-load test, and a configuration is blind exactly when a uniformly loaded structure shows no deflection at the sensor. The invariance has a simple reading: each axle adds a shifted, load-scaled copy of the same influence line, and a translation changes the shape of the record but not its signed area. C_m is a global flexibility seen from the sensor; for a continuous beam it is a signed sum, and its sign governs the results below. The property requires linearity and time-invariance; it does not survive material or geometric nonlinearity, bearing uplift or friction, or a stiffness change during the crossing.

For a strain sensor the same argument applies to the bending moment: the strain integral over a crossing equals the total load times $c\,M_q(x_m)/(E\,I\,q_0)$, where M_q is the bending moment under the uniform load and c the distance of the fibre from the neutral axis, so a strain sensor is blind at the contraflexure points of the uniform-load moment diagram (for example $0.75\,L$ from the outer support of an equal two-span beam). Such points exist on every continuous girder, whereas the deflection blind configurations derived below exist only for particular span and stiffness ratios.

2.2. Two-span continuous beam

Let the instrumented span have length L_1 and rigidity $E\,I_1$, the adjacent span $L_2 = \rho_L L_1$ and $E\,I_2 = \rho_{EI} E\,I_1$, and the sensor be at $\xi = x_m / L_1$ (Fig. 1a). Slope continuity at the interior support under a uniform load gives (Appendix A)

$$C_m = \frac{L_1^4}{24\,E\,I_1} F(\xi, \mu), \quad F(\xi, \mu) = \xi\left(1 - 2\xi^2 + \xi^3\right) - \frac{\mu}{2}\xi\left(1 - \xi^2\right), \tag{4}$$

$$\mu = \frac{1 + \rho_L^3/\rho_{EI}}{1 + \rho_L/\rho_{EI}}. \tag{5}$$

Geometry and stiffness enter only through the scalar μ, which is unity for a symmetric beam. For equal spans and a mid-span sensor Eq. (4) gives $C_m = L^4/(192\,EI)$. Table 1 compares Eq. (4) with a finite element (FE) evaluation of Eq. (3); the two agree to eleven significant figures.

Table 1. Two-span beam: closed form, Eq. (4), against FE evaluation of Eq. (3). Values are $192\,EI_1 C_m/L_1^4$; $\rho_{EI} = 1$.

Case	ρ_L	ξ	Closed form	FE	Difference
Equal spans	1.0	0.50	1.00000000	1.00000000	1×10^{-11}
Short adjacent span	0.5	0.50	1.37500000	1.37500000	1×10^{-11}
Long adjacent span	1.3	0.50	0.41500000	0.41500000	9×10^{-11}
Blind configuration	1.45743	0.50	0.00000000	0.00000000	7×10^{-11}
Beyond blind	2.0	0.50	−2.00000000	−2.00000000	8×10^{-12}
Optimal sensor	1.0	0.42154	1.03989535	1.03989535	9×10^{-11}
Sensor near support	1.0	0.70	0.60480000	0.60480000	1×10^{-11}

Because both estimators of Section 2.4 divide by C_m, the configurations with $C_m = 0$ carry no information about the load. Setting $F = 0$ gives a condition on μ alone, which inverts for the stiffness ratio:

$$C_m = 0 \Leftrightarrow \mu = \mu_{crit}(\xi) = \frac{2\left(1 + \xi - \xi^2\right)}{1 + \xi}, \rho_{EI,crit} = \frac{\rho_L\left(\rho_L^2 - \mu_{crit}\right)}{\mu_{crit} - 1}. \tag{6}$$

For a mid-span sensor $\mu_{crit} = 5/3$. With uniform stiffness the blind condition is $3\rho_L^2 - 3\rho_L - 2 = 0$, and maximising C_m over ξ at $\mu = 1$ gives $8\xi^2 - \xi - 1 = 0$. Both roots are exact:

$$\rho_{L,crit} = \frac{3 + \sqrt{33}}{6} = 1.45743, \xi^{\iota} = \frac{1 + \sqrt{33}}{16} = 0.42154. \tag{7}$$

Beyond $\rho_{L,crit}$ the constant changes sign but remains observable, because the estimators need only $C_m \neq 0$: at $\rho_L = 2$, $\left|192\,E\,I_1\,C_m / L_1^4\right| = 2.0$, twice the equal-span value. The response is then dominated by the uninstrumented span, so the constant depends mainly on the rigidity and supports of that span, and the record is processed with its sign reversed. Observability is lost only at the blind configuration and degrades in a band on both sides of it. At ξ¿ the constant exceeds its mid-span value only by the factor 1.039895, whereas it falls to 10 % of its maximum at $0.9\,L_1$.

The degeneracy is driven almost entirely by the adjacent span: its negative contribution grows much faster with ρ_L than the positive contribution of the instrumented span, and the same decomposition places the optimum off mid-span (Appendix A).

Since $\partial\mu / \partial\rho_{EI} \propto \rho_L\left(1 - \rho_L^2\right)$, stiffening a long adjacent span raises C_m and stiffening a short one lowers it. At $\rho_L = 1.6$ the constant rises from −0.78 at $\rho_{EI} = 0.5$ to +0.33 at $\rho_{EI} = 4$, crossing zero at $\rho_{EI} = 2.14$. Blindness belongs to the pair (configuration, sensor position), not to the structure: at $\rho_L = 1.45$ a mid-span sensor gives 0.021, whereas a sensor at $\xi = 0.266$ gives 0.233. Fig. 2 maps the constant over the two ratios with the blind locus of Eq. (6) and the near-blind band $\left|192\,E\,I_1\,C_m / L_1^4\right| < 0.2$, where the measurement sensitivity exceeds five times its equal-span value. For a mid-span sensor and uniform stiffness this band spans $1.385 < \rho_L < 1.525$, on both sides of the blind ratio.

2.3. Three-span continuous beam

Eq. (3) extends without new ingredients to the symmetric three-span layout with a long main span that is usual for medium-span box girders; only the number of redundants increases. Let the side spans have length L_s and rigidity $E\,I_s$, and the main span $L_m = \beta\,L_s$ and $E\,I_m = \rho_{EI}\,E\,I_s$, with $E\,I$ piecewise constant (Fig. 1b). For a side-span sensor, ρ_{EI} is again the adjacent-to-instrumented stiffness ratio. By symmetry the two support moments are equal, and the three-moment equation gives $M_B = M_C = -\,q_0\,L_s^2\,\mu / 8$. For a sensor at $\xi = x / L_s$ in a side span, C_m has the form of Eq. (4) with

$$\frac{192\,E\,I_s\,C_m}{L_s^4} = 8\,F\left(\xi, \mu\right), \mu = \frac{2\left(\rho_{EI} + \beta^3\right)}{2\,\rho_{EI} + 3\,\beta}. \tag{8}$$

Because the far side span restrains the interior support, three equal spans give $\mu = 4/5$ rather than 1 and a mid side-span constant 1.3 times the two-span value. The main-span counterpart, with the interior-moment parameter ν, is given in Appendix D; both agree with FE evaluations of Eq. (3) to round-off.

Because F is unchanged, the blind condition for a side-span sensor is again $\mu = \mu_{crit}(\xi)$ of Eq. (6). At mid side span the blind stiffness ratio is explicit, $\rho_{EI,crit} = 3\beta(2\beta^2 - 5)/4$. For uniform rigidity the critical main-to-side span ratio is the real root of a cubic in the irreducible case, so its exact closed form is trigonometric, Eq. (9):

$$6\beta^3 - 15\beta - 4 = 0,\ \beta_{crit} = \sqrt{10/3}\cos\left[1/3\arccos\left(2/5\sqrt{6/5}\right)\right] = 1.700594. \tag{9}$$

The near-blind band $\left|192\,E\,I_s C_m / L_s^4\right| < 0.2$ at mid side span spans $1.621399 < \beta < 1.775$; its lower edge is the root of $30\beta^3 - 69\beta - 16 = 0$. The value $\beta = 2\cos(\pi/9) = 1.879385$ ends the side-span blind locus: for larger β every side-span position has a negative, non-zero constant dominated by the main-span load, as beyond the two-span blind ratio. At $\beta = 2$ the mid side-span value is $\left|192\,E\,I_s C_m / L_s^4\right| = 0.875$, outside the near-blind band. The ratio of the negative main-span contribution to the positive side-span contributions, for a mid side-span sensor and uniform rigidity, is

$$\frac{\left|C^{-}\right)}{C^{+\text{¿}} = \frac{6\beta^3}{15\beta + 4}.\text{¿}} \tag{10}$$

The optimal side-span position solves $8\xi^3 - (12 - 3\mu)\xi^2 + (2 - \mu) = 0$. As Table 2 shows, $\xi^{\text{¿}}$ moves towards the abutment as the main span lengthens. A mid main-span sensor is blind at $\beta = 0.964706$, with a main span slightly shorter than the side spans, and has a negative constant for smaller β. Since $\partial\mu/\partial\rho_{EI} \propto 3\beta - 2\beta^3$, for $\beta > \sqrt{3/2}$ a stiffer main span lowers μ and improves the observability of the side span. Haunched girders, however, are stiff near the piers rather than uniformly stiffer in the main span, and Section 4 shows that their behaviour is not captured by a piecewise-constant ρ_{EI}.

Table 2. Symmetric three-span beam, uniform rigidity: optimal side-span sensor position and the indicator $192\,E\,I_s C_m / L_s^4$. The FE slope at $\xi^{\text{¿}}$ is at most 3.6×10^{-7} in every case.

β	μ	$\xi^{¿}$	At $\xi^{¿}$	At $\xi=1/2$
1.0	0.80000000	0.44603660	1.32176895	1.30000000
1.2	0.97428571	0.42508293	1.07563406	1.03857143
1.4	1.20774194	0.38747142	0.75849895	0.68838710
1.5	1.34615385	0.35820014	0.58065335	0.48076923
1.63636	1.55785124	0.29944923	0.33162780	0.16322314
1.7	1.66563380	0.26077470	0.22018607	0.00154930

Fig. 3a maps the indicator over β and ξ, with the blind locus, the near-blind band and $\xi^{¿}(\beta)$. Fig. 3b treats asymmetric layouts with the general closed form (Appendix D). The far side span matters little while it is short: for L_3/L_2 between 0.3 and 0.6, the blind ratio L_1/L_2 for a mid span-1 sensor stays between 0.587 and 0.592.

2.4. Estimators and observability indicators

If E, I and the spans are known, Eq. (2) gives the beam-property method (BPM), $P_{tot}=I_y/C_m$. Because $C_m \propto L^4$, a 1 % error in span length produces a 4 % error in the load, and on an existing bridge EI is rarely known to better than a few per cent. The standard-load method (SLM) removes this requirement: a load of known total weight P_0 crosses the structure, and since both crossings share C_m,

$$P_t = P_0 \frac{I_{y,t}}{I_{y,0}}. \tag{11}$$

Eq. (11) is the influence-area calibration of Ojio and Yamada [27] written for a deflection record. The procedure is: (i) weigh a reference load; (ii) record one or, preferably, several complete crossings of it and average the integrals; (iii) record the unknown crossing; (iv) evaluate both integrals with the same quadrature rule and sampling grid; (v) apply Eq. (11). SLM cancels the structural factor shared by the two crossings, and with it the dependence on flexural rigidity, span length and support conditions. Other systematic errors cancel only to the extent that they are common to the two crossings, and independent random errors do not cancel at all (Section 6).

Records are discrete. With samples y_i at $x_i = i\,\Delta x$ the trapezoidal and Simpson rules are

$$I_y^{trap} = \Delta x\left[y_0/2 + \sum_{i=1}^{n-1} y_i + y_n/2\right], I_y^{Simp} = \frac{\Delta x}{3}\left[y_0 + 4\sum_{i\,odd} \square\, y_i + 2\sum_{i\,even\neq 0,n} \square\, y_i + y_n\right]. \tag{12}$$

The Simpson rule of Eq. (12) needs an even number of intervals n; for odd n the last three intervals use Simpson's 3/8 rule. The end half-weights vanish only for a record that starts and ends at zero deflection, as in the laboratory but not on a bridge with residual deflection. Because the influence line of a continuous beam is piecewise cubic and twice differentiable, Simpson quadrature is nearly exact for a single load, whereas the trapezoidal rule converges only as Δx^2 (Section 3.2). Each further axle, however, adds slope discontinuities where it enters and leaves the beam, and Simpson quadrature keeps this accuracy only when these kinks fall on panel edges; otherwise the trapezoidal rule with $\Delta x \le l/20$ is the safer choice.

Both estimators divide by C_m, so to first order the measurement sensitivity scales as $1/|C_m|$. We therefore propose $|C_m|$, in the normalised form

$$N_m = \frac{192\, E\, I_{ref}\, C_m}{L^4}, \tag{13}$$

as a pre-installation observability indicator, with L the span containing the sensor. It answers three questions at once: whether a span arrangement can observe the load from a given point, where to place the sensor, and how tight the baseline stability must be (Section 6.2). The threshold $|N_m| < 0.2$ of Eq. (13) marks near-blind locations for prismatic beams.

For non-prismatic girders N_m depends on the choice of $E\, I_{ref}$, and Section 4 shows that this choice can dominate the result. A complementary measure that does not need a reference stiffness is the cancellation ratio

$$R_m = \frac{C_m}{A^+ = 1 - \frac{|A^-|}{A^+, ¿}¿} \tag{14}$$

the net influence area divided by its positive part. $R_m = 1$ for a simply supported beam, $R_m = 4/7$ for an equal two-span beam instrumented at mid-span, $R_m = 0$ at a blind configuration and $R_m < 0$ beyond it. N_m measures the signal relative to the local flexibility; R_m measures how much of it the adjacent spans cancel. No threshold is proposed for R_m; it is used to rank candidate locations by $|R_m|$. Neither index carries the absolute signal level, on which the offset error depends; the offset-equivalent load of Section 6.2 supplies it.

A third index states what invariance costs. The integral weights the record uniformly, the only weight invariant to translation, whereas under white noise and with the structure and axle positions known the optimal weight would be the influence line itself (a matched filter). By the Cauchy–Schwarz inequality the ratio of the standard deviations of the two estimators is

$$O_m = \frac{\left|C_m\right)}{\sqrt{L_{tot}\int_0^L \eta_m^2\, d\,x}} \le 1. \tag{15}$$

For the equal two-span beam instrumented at mid-span, Eq. (15) gives $O_m = 2\sqrt{44765}/1279 = 0.331$. The integral estimator keeps one third of the precision of an estimator that knows the structure and the axle positions, and in exchange needs neither. O_m decreases monotonically as the sensor moves away from the outer support, whereas $\left|C_m\right)$ peaks at ξ'; sensor placement should follow $\left|C_m\right)$.

The closed forms of Eqs. (4)–(10) are exact for prismatic or piecewise-prismatic beams. The next two sections test the two-span result on a laboratory beam and apply the indicators to a haunched three-span girder.

3. Laboratory verification

This section verifies the influence line, the integral and the two estimators on a laboratory two-span beam. The loads were positioned statically at successive points along the beam. The tests therefore verify the model and the algorithm, not performance under moving traffic.

Every theoretical claim tested here is a dimensionless identity. Eq. (2) holds for any linear-elastic beam, and the ratios of measured to predicted integral and of identified to weighed load do not depend on the size of the beam; the blind conditions are statements about ρ_L, β and ξ alone. The laboratory beam therefore tests the identities and the estimators, not the scale. The quantities that do depend on scale, the offset sensitivity L_{tot}/I_y and the ratio T_{cr}/T_1, are evaluated for the in-service bridge and for bridge spans in Sections 6.2 and 6.5.

3.1. Specimen and procedure

Tests were carried out at the Laboratory of Transportation Technology and Systems, Faculty of Bridge and Road Engineering, Hanoi University of Civil Engineering. The specimen (Fig. 4) was a steel beam of total length 5400 mm on three supports forming two equal spans of

2500 mm; the remaining 400 mm of overhang beyond the end supports is neglected in the analytical influence line. The end supports were steel rollers and the interior support a steel knife edge that restrains vertical displacement and allows rotation. The cross-section was a hollow square tube, 25 mm × 25 mm with a wall thickness of 2.03 mm (averaged from measurements), giving $I = 16\,529.8$ mm⁴. With $E = 200$ GPa, $EI = 3.3060 \times 10^9$ N mm² and the theoretical constant at mid-span of the first span is $C_m = l^4/(192\,EI) = 61.541$ mm²/N. Shear deformation contributes 8×10^{-4} of the bending deflection (span-to-depth ratio 100), so Euler–Bernoulli theory applies.

Deflection at mid-span of the first span was measured with a linear variable differential transformer (LVDT) of range 0–50 mm and resolution 0.005 mm, connected to a TDS-303 data logger. Calibrated steel weights were suspended at positions 250 mm apart along the whole traverse, so $\Delta x = 250$ mm. Each reported deflection is the mean of three loading and unloading cycles. Three configurations were tested: a single load of 4.143 kg including its hook (40.64 N), used as the reference; two loads of 2145 g and 4684 g at 250 mm spacing; and three loads of 2550 g, 5100 g and 5100 g, with the 2550 g load leading, the second load 500 mm behind it and the third a further 250 mm behind. This order is the one that fits the record of Appendix C (root-mean-square residual 0.030 mm, against 0.55 mm for the reverse spacing). Each record extends to $2l + a_{max}$, that is 5000, 5250 and 5750 mm. The beam was modelled in Abaqus [45] with beam elements; cubic elements reproduce the exact nodal Euler–Bernoulli displacements, so agreement with the analytical influence line verifies the implementation only. The FE model was used to verify Eq. (3) and for the parametric studies.

3.2. Influence line and integral

Fig. 5 compares the analytical influence line (Appendix A, Eq. (A.2)) with the measurements for the single load; the values are listed in Appendix C. The largest absolute deviation is 0.0126 mm, which is 0.44 % of the maximum deflection and two and a half times the LVDT resolution. The residuals are predominantly positive on both spans, which indicates a small systematic component, plausibly slight support flexibility or a small overestimate of EI.

Table 3. Deflection integral for the single-load configuration.

Quantity	Value (mm²)	Difference from exact
Exact, $P_0 l^4/(192\,EI)$	2501.19	n/a
Analytical influence line, trapezoidal rule, Δx = 250 mm	2488.68	−0.500 %

Quantity	Value (mm²)	Difference from exact
Analytical influence line, Simpson rule, Δx = 250 mm	2501.85	+0.027 %
Measured, trapezoidal rule, Δx = 250 mm	2513.43	+0.49 %

In Table 3, the trapezoidal bias of −0.500 % is a property of the quadrature, not an experimental error, and it propagates one-for-one into BPM. Simpson quadrature on the same grid reduces it to +0.027 % at no experimental cost. The single-load record has 20 intervals; the two- and three-load records have 21 and 23, so Simpson quadrature there needs the 3/8 end panel of Section 2.4. It then loses its advantage: on the analytical records its bias is −0.92 % for the two-load and −0.44 % for the three-load group, against −0.500 % for the trapezoidal rule, because the entry and exit kinks of the trailing loads fall inside Simpson panels.

3.3. Total-load identification

The measured records of the two- and three-load groups (Appendix C) were used to identify the total load by BPM and by SLM, with the single-load record as reference. Table 4 compares the results with direct weighing.

Table 4. Identified total loads against direct weighing.

Configuration	Method	I_y (mm²)	Identified load (N)	Deviation
Two loads	Weighing	n/a	66.99	n/a
	BPM	4168.15	67.73	+1.10 %
	SLM	4168.15	67.40	+0.61 %
Three loads	Weighing	n/a	125.08	n/a
	BPM	7788.93	126.57	+1.19 %
	SLM	7788.93	125.95	+0.70 %

Both methods identify the total load within 1.2 %, and SLM is the more accurate. The ratio of the two estimates is 1.00489 for both configurations and equals the ratio of the measured reference integral to its exact value, 2513.43/2501.19. The whole difference between BPM and SLM is thus the calibration error of the reference record, which SLM absorbs and BPM does not. The BPM result also benefits from two errors of opposite sign: the trapezoidal rule underestimates the integral by 0.50 %, while the measured deflections exceed theory. Correcting the quadrature bias alone would raise the BPM deviation from +1.10 % to +1.61 %.

The excess of the measured over the analytical integral, both by the trapezoidal rule on the same grid, depends on the configuration: the ratio is 1.010 for the single load (40.64 N), 1.016

for the two-load group (66.99 N) and 1.017 for the three-load group (125.08 N). SLM divides by the single-load record, so the step from 1.010 to 1.016–1.017 is exactly its residual of 0.61 % and 0.70 %. The present data cannot separate the possible causes: a mild non-proportionality of the support seating, an error in the stated masses, or errors in the positions of the suspended loads would each produce such a step. Linearity is the premise of both Eq. (2) and Eq. (11), so the direct invariance test recommended in Section 7 should repeat each spacing with loads of different magnitude.

Fig. 6 plots the measured integral against the total load for the three configurations. A regression through the origin gives $I_y = 62.229\,P$ with $R^2 = 0.99998$, a slope 1.12 % above the theoretical 61.541 mm²/N, which is consistent with the residual pattern of Fig. 5. Because the three configurations differ in both total load and spacing, this line supports the invariance only indirectly; three points do not make the R^2 a strong test. A direct test with a fixed total load at varying spacings is recommended in Section 7.

The laboratory beam thus confirms the influence line, the integral and the common-mode cancellation of SLM on the equal two-span case. The next section asks what the criterion says about a real continuous girder whose layout lies close to the blind locus.

4. Application to an in-service three-span box girder

This section applies the observability criterion to the acceptance-test data of an in-service three-span box girder. The bridge was tested only with parked convoys, one reading per gauge and load case, so the data cannot check Eq. (2) itself. They can show how far the observability of each installed gauge is determined by the design data and two static readings.

4.1. Bridge and test data

The Cầu Mây Bridge carries provincial road DT389 over the Kinh Mon River in Hai Duong, Viet Nam. Its main bridge is a three-span continuous prestressed concrete box girder of 55 + 90 + 55 m, built by balanced cantilever, so $\beta = 90/55 = 1.636$. The depth varies from 2.40 m at mid-span to 5.40 m at the piers. The designer's rating calculation gives $I = 5.482$ m⁴ at mid-span and $I = 59.612$ m⁴ at the pier face for vertical bending, with $E_c = 36\,057$ MPa ($f'_c = 45$ MPa). The inner piers T6 and T7 are monolithic with the girder, and the end piers T5 and T8 carry pot bearings. The approach spans are simply supported behind expansion joints and add nothing to C_m.

The acceptance load test was carried out in June 2020 by daylight, in hot and sunny weather. The convoys on the continuous spans were made up from ten three-axle trucks, nominally 5 + 10 + 10 t. All ten were weighed, with a mean gross mass of 25.06 t; the masses of the individual trucks in each convoy are not used here. The axle spacings were 3.60 m and 1.35 m and the track width 1.85 m. Trucks stood two abreast, 1.1 m apart, with a clear gap of 6.0 m between consecutive trucks along the span. Scheme III placed four trucks (100 t nominal) in side span N8, with the middle axle of the first row at 0.4L. Scheme IV placed six trucks (150 t nominal) in centre span N7, the middle axle of the second row at mid-span. The truck positions were set out with a tape measure and marked with paint, and the positioning error is estimated at less than 20 cm in each direction. Deflections were read with Mitutoyo dial gauges, three per section, under centric and eccentric transverse positions. In the eccentric position the outer wheel was 0.6 m from the kerb. Each scheme was loaded three times with zero readings in between, and the tables give one peak value per gauge and load case. Gauge N8 is at 0.4L, 22 m from the end support; gauge N7 is at mid-span of the centre span. The mean of the six readings is 5.55 mm at N8 against 5.93 mm from the designer's FE model, and 15.42 mm at N7 against 16.70 mm.

4.2. Model and calibration

Influence lines and C_m were computed with a two-dimensional Euler–Bernoulli FE model with variable stiffness, 10 elements per metre and rigid vertical supports. The depth was assumed to vary parabolically over a length L_v from each inner pier axis, and $I = I_{mid}\left(h/h_{mid}\right)^n$ with $n = 2.943$ fitted to the two known sections (Fig. 7a). L_v is not stated in the available documents; $L_v = 44$ m was used, bracketed by 40 m and 45 m. Each inner pier was represented by a rotational spring k. The section properties were checked independently: a polygon routine applied to the cross-section drawings reproduces the designer's areas within −0.3 % and −0.5 % and the second moments within −5.8 % and −2.6 % at mid-span and at the pier face. Shear deformation, shear lag and transverse distribution are not modelled; the beam carries the total load of the two trucks in each row.

The pier dimensions were not available, so k had to be calibrated, and the static data admit two calibrations. Calibration A keeps the design rigidity and fits k to the deflections that the test report gives for the designer's FE model: $k_{FE} = 2.7 \times 10^7$ kN m/rad ($2.5–3.6 \times 10^7$ over the L_v variants), which reproduces both report values within −4.2 % to +3.7 %. Scaling the rigidity uniformly at this k to match the measured deflections gives

$\alpha = EI_{eff}/EI_{design} = 1.088$, with misfits of −4.6 % and +5.0 %. Calibration B fits α and k jointly to the measured deflections and gives $\alpha = 0.716$ and $k = 1.2 \times 10^{8}$ kN m/rad, with misfits of −2.8 % and +2.9 %. Both sets of misfits are smaller than the spread of the individual readings (5.22–5.91 mm at N8, 14.62–16.67 mm at N7). The calibrations use the nominal axle loads. Replacing them by the weighed mean of 25.06 t, with every axle scaled by the same factor, changes the measured-to-model ratios of Table 6 by at most 0.003 and the indicators of Table 5 by at most 0.002, whether or not the report FE values are assumed to use the weighed mass; these changes are below the printed precision of the conclusions and the nominal-load values are kept. Two static readings cannot separate girder rigidity from pier stiffness: a stiffer girder on more flexible piers and a softer girder on stiffer piers deflect alike at the two gauges. The pier drawings, the designer's FE model or measured natural frequencies would be needed to choose between the two calibrations, and none was available.

The indicator N_m of Eq. (13) uses $EI_{ref} = EI_{mid}$ of the design section, the minimum of $EI(x)$, in both calibrations, so that the two are compared on one scale. As a check, C_m computed as the influence-line area and as the deflection under a unit uniform load, Eq. (3), agreed to 1.1 × 10^{-4} in every case. The difference is the error of the trapezoidal rule applied to the nodal influence-line values on the 0.1 m mesh; the uniform-load route uses consistent nodal loads and has no such error.

4.3. Observability of the installed gauges

The prismatic closed form already places this layout near the blind locus. With uniform rigidity, Eq. (8) gives $N_m = 0.163$ at mid side span, inside the near-blind band, because $\beta = 1.636$ lies between the band edge 1.621399 and β_{crit} of Eq. (9). Eq. (10) gives $|C^{-}|/C^{+} = 0.921$. Moving the sensor to $\xi = 0.2994$ would double the indicator to 0.332 (Table 2). The span-wise contributions, obtained by loading one span at a time, are 1.870395 and 0.195395 for the two side spans and −1.902566 for the main span, in units of $L_s^4/(192\,EI_s)$; the influence line keeps one sign in each span (+, −, +). A mid main-span sensor of the same prismatic beam would read 0.754630 with a cancellation of only 0.30.

Table 5 gives the results of the variable-stiffness model for both calibrations, and Figs. 7b and 7c show the influence lines.

Table 5. Observability indicators for the Cầu Mây gauges. $N_m = 192\,EI_{mid} C_m / L^4$, Eq. (13), with L the span containing the gauge and EI_{mid} of the design section; $R_m = C_m / A^{+¿¿}$, Eq. (14). Calibration A: design EI, $k = k_{FE}$. Calibration B: $\alpha = 0.716$, $k = 1.2 \times 10^8$ kN m/rad. The last column covers pinned to fixed piers with design EI and all L_v variants; these limits are bounds, not an uncertainty range.

Gauge	Constant EI, pinned: N_m	A: N_m	A: R_m	B: N_m	B: R_m	Pinned … fixed bound of N_m
N8, 0.4L of side span (installed)	0.287	0.181	0.18	0.476	0.56	0.023 … 0.446
N8 at mid side span (alternative)	0.163	0.105	0.11	0.377	0.49	−0.066 … 0.375
N7, mid centre span (installed)	0.755	0.209	0.59	0.234	0.77	0.145 … 0.263

The observability of the side-span gauge cannot be identified from the design data and the two static readings. Under calibration A its indicator is 0.181 and the negative lobe in the 90 m centre span cancels 82 % of its positive influence area. Scaling the rigidity by $\alpha = 1.088$ at k_{FE} changes these values only to 0.160 and 83 %. Under calibration B the indicator is 0.476 and the cancellation 44 %. Both calibrations reproduce the measured deflections, so the data allow both a poorly and a reasonably conditioned gauge. The same holds locally around calibration A: halving or doubling k moves the N8 indicator from 0.124 to 0.248, and its constant varies as $k^{0.51}$ ($\partial \ln C_m / \partial \ln k = 0.51$ at k_{FE}). The pinned and fixed limits bracket the indicator between 0.02 and 0.45; they are not realistic states of monolithic piers and are quoted only as bounds. For the alternative mid side-span position, which the test tables use as a label for N8 while the test plan places the gauge at 0.4L, the bracket runs from −0.066 to 0.375 and contains an exactly blind state.

The centre-span gauge is stable. Its indicator is 0.209 under calibration A and 0.234 under calibration B, and halving or doubling k around k_{FE} keeps it between 0.191 and 0.224 ($\partial \ln C_m / \partial \ln k = -0.12$). The two gauges differ because the net area at N8 is the difference of two lobes of comparable size, and the pier stiffness controls how much of the centre-span lobe reaches the side span; at N7 the net area is dominated by the instrumented span itself. A side-span gauge on a continuous box girder is therefore ill-conditioned with respect to pier rotational stiffness, while a centre-span gauge is not.

Including the haunches lowers the centre-span indicator from 0.755 for a prismatic beam to 0.21–0.23. The cause is not cancellation, since R_m is 0.59–0.77: the haunches restrain the

span, so the net area is small relative to the mid-span flexibility used for normalisation. The threshold $|N_m|<0.2$ derived for prismatic beams therefore does not transfer directly to haunched girders; it must be stated together with EI_{ref}. The cancellation ratio separates the two effects, and by R_m N7 is the better conditioned gauge under both calibrations. The absolute constants point the same way: C_m is 3.61×10^{-4} m²/kN at N7 and 4.38×10^{-5} m²/kN at N8 under calibration A, and 4.05×10^{-4} and 1.15×10^{-4} m²/kN under calibration B. For the same vehicle the centre-span gauge records an integral 8.2 or 3.5 times larger, and Section 6.2 converts this into the error caused by a residual baseline offset.

4.4. Static consistency check and implications for BPM and SLM

Table 6. Static consistency check. Measured values are the mean of three gauges under two transverse positions. Model values use the nominal axle loads 5 + 10 + 10 t.

Gauge	Convoy	Measured (mm)	Report FE (mm)	Model A (mm)	Model B (mm)	Measured / report FE	Measured / model A	Measured / model B	Pinned … fixed (mm)
N8, 0.4L	4 × 25 t	5.55	5.93	5.69	5.40	0.936	0.977	1.029	7.18 … 2.75
N7, mid	6 × 25 t	15.42	16.70	17.32	15.87	0.923	0.890	0.972	22.99 … 8.34

The measured deflections are 0.936 and 0.923 of the designer's model at the two gauges. Against model A the ratios are 0.977 and 0.890; a single rigidity factor cannot remove this difference between the gauges, which is why calibration A matches the measurements only to −4.6 % and +5.0 % after scaling by $\alpha = 1.088$. Against model B the ratios are 1.029 and 0.972. The effective girder rigidity is thus 1.088 or 0.716 times the design value depending on how the pier stiffness is resolved, and the static data do not decide between them. The pinned and fixed limits bracket the measurements widely (Table 6), which shows how much the support condition alone controls the response.

Linearity, on which both the theorem and SLM rest, cannot be tested directly: no span was loaded with more than one convoy size while deflections were read. The evidence is indirect. The measured-to-model ratios are similar for the four-truck and six-truck convoys (0.936 and 0.923). Transverse effects are visible: individual readings range from 5.22 to 5.91 mm at N8 and from 14.62 to 16.67 mm at N7, and in the eccentric N7 case the readings across the section spread by up to 2.05 mm. A single-gauge installation would carry the transverse position of the vehicle as an error term, which a one-dimensional treatment cannot remove. The longitudinal positioning error of the parked trucks is small by comparison: shifting both

convoys by 0.20 m along the span changes the model deflection by at most 0.6 % at N8 and by 0.01 % at N7. The effect of the transverse positioning error, and of temperature changes between the zero and the loaded readings during the daytime tests in hot weather, could not be quantified from the available records.

These results have three practical consequences. First, a C_m taken from the design model is not usable at the side-span gauge: the data admit constants that differ by a factor of 2.6 between the two calibrations, and the design model gives no basis for choosing one. Second, the centre-span gauge is robust; its constant differs by 12 % between the calibrations, so a design-model value would carry an error of that order. Third, SLM calibrates C_m with a weighed vehicle and removes the identifiability problem, provided the response is linear. It does not remove the residual-offset term, which at the side-span gauge remains large (Section 6.2). For a side-span gauge on a continuous girder, C_m must therefore come from SLM or from a model identified in situ, for example from measured influence lines [43], and never from the design model. The criterion is most useful before installation: on this bridge a gauge in the centre span, or a second gauge there, would have avoided the ill-conditioned side-span location.

The in-service girder thus indicates, in a model built from the designer's section data, that a side-span gauge on a layout near the prismatic blind locus is ill-conditioned with respect to pier stiffness, and that the cancellation ratio separates true cancellation from the effect of haunching. A direct test of Eq. (2) on a continuous girder still requires crawl or moving passes of weighed vehicles.

5. Field deflection records: baseline drift and filtering

This section has a supporting role: moving-truck deflection records from two simply supported bridges illustrate how baseline drift and filtering affect the integral (Section 6), not the absolute relation of Eq. (2). Truck weight, speed and lane were not recorded, the channel calibration is not documented (the ratio between the processed and the raw channel differs by span, from 0.67 to 5.15), and the series are quantised (0.1 mm steps in the raw Gie Cuc channels) and partly processed by hand.

The records come from Gie Cuc I (three simply supported spans of 25.7 m) and Phuong Do (two spans, calculation length 17.5 m), measured at mid-span with an SDA-830C recorder at 200 Hz for 15 s; seven complete crossings were identified. Each crossing integral was

computed with four baselines: a constant tare level, a straight line between quiet windows before and after the event (the reference), a cubic fitted to all quiet samples, and zero-phase high-pass filters at 0.1 Hz and 0.5 Hz (Fig. 8). The residual offset after the linear pre/post correction cannot be resolved from these data: the post-minus-pre level difference is 0.0–0.1 mm at Gie Cuc, at the 0.1 mm resolution of the raw channels, and 0.17–0.23 mm at Phuong Do N2, and the quiet-level scatter is 0.01–0.13 mm. Residual offsets of a few hundredths of a millimetre, as assumed in Section 6, were therefore not demonstrated on these channels.

Over the 2–4 s event window the tare, linear and cubic baselines agree within about 6 % for every crossing but one; the cubic differs from the linear by +1.7 to +3.6 % at Gie Cuc and −1.2 to +0.1 % at Phuong Do. The exception is the raw Phuong Do N1 record, whose baseline steps by 2.06 mm (18 % of the peak) as the truck leaves the span; there the tare-only integral is 36.4 % below the reference. The effect of a small offset grows with the integration time: over the whole 15 s record, tare-only integrals differ from the cubic-baseline values by +7 %, −16 % and −60 % at the three Gie Cuc spans, +34 % at Phuong Do N2 and −170 % (a sign reversal) at Phuong Do N1, which is the behaviour of Eq. (19) in Section 6.2. High-pass filtering changed the event integral by −44 to −83 % at 0.1 Hz and by about −100 % at 0.5 Hz; a filter with zero gain at zero frequency removes the quantity being integrated and cannot serve as a baseline correction.

A zero-phase low-pass filter at 2 Hz, about one third of the measured fundamental frequencies of 5.2–6.0 Hz, left the event integral unchanged within a ratio of 0.9999 to 1.0010 for complete events (Fig. 9). The dynamic residual, unfiltered minus filtered, integrates to 0.001–0.022 of its absolute integral, although its peaks reach 3–16 % of the quasi-static peak. This partly follows from the unit gain of the filter at zero frequency and illustrates the processing rather than the theory. Fitted speeds and speed-normalised integrals cannot be interpreted with the undocumented channel scales and are given in the supplementary material.

The records thus support two operational rules: re-zero the baseline in a window tight around the crossing, and remove dynamics with a low-pass, never a high-pass, filter.

6. Error budget and dynamic validity

The laboratory agreement of Section 3 was obtained on a beam of known properties under static loads. This section develops the error budget explicitly, evaluates the offset term at the

scale of the bridge of Section 4 and bounds the quasi-static assumption, so that the results can be extrapolated with stated sensitivities. Monte Carlo details are in Appendix B.

6.1. Measurement model and error propagation

The recorded deflection at sample i is modelled as

$$y_i^{meas} = (1+\gamma)\, y_i + \varepsilon_0 + \sigma_a \zeta_i, \zeta_i \sim N(0,1), \tag{16}$$

with gain error γ, zero offset ε_0 and additive noise of standard deviation σ_a. Here ε_0 is the offset that remains after the baseline has been re-zeroed before and after the crossing, not the absolute drift of the sensor. Unlike a purely multiplicative model, Eq. (16) does not predict zero noise at the supports. To first order, the error of each estimator is given by Eqs. (17) and (18):

$$\left(\frac{\delta P}{P}\right)_{BPM} = \frac{\delta(EI)}{EI} - 4\frac{\delta l}{l} + \gamma_t + e_{q,t} + \frac{\varepsilon_{0,t} L_t}{I_{y,t}} + \frac{\delta v_t}{v_t}, \tag{17}$$

$$\left(\frac{\delta P}{P}\right)_{SLM} = \frac{\delta P_0}{P_0} + (\gamma_t - \gamma_0) + (e_{q,t} - e_{q,0}) + \left(\frac{\varepsilon_{0,t} L_t}{I_{y,t}} - \frac{\varepsilon_{0,0} L_0}{I_{y,0}}\right) + \left(\frac{\delta v_t}{v_t} - \frac{\delta v_0}{v_0}\right), \tag{18}$$

where e_q is the quadrature bias, subscript t the target crossing and subscript 0 the reference. SLM removes the structural terms of Eq. (17) outright, because they belong to C_m. Every other term survives in Eq. (18) as a difference, which vanishes only when the error is common to both crossings. A gain error cancels if the chain is not recalibrated between the crossings. The quadrature bias cancelled in the present tests (−0.5000 % for all three configurations) only because the spacings were integer multiples of Δx; for the two-load group at a spacing of 300 mm the bias would be −0.171 %, at 375 mm +0.014 %, and the residual would reach 0.33 and 0.51 percentage points. The zero-offset term never cancels fully, even for equal offsets, because the sensitivity L/I_y differs: for the two-load case the residual is $\varepsilon_0(1.2596 - 1.9893)$ mm^{-1}, so its magnitude is 58 % of the BPM offset term, a 42 % reduction. A reference load as heavy as the target vehicles makes the cancellation nearly complete.

6.2. Residual zero offset

A constant offset contributes $\varepsilon_0 L_{tot}$ to the integral without any cancellation, whereas the net integral is itself the residue of the cancellation between spans. With $I_y = C_m P$, the relative bias is

$$\left(\frac{\delta P}{P}\right)_{\varepsilon_0} = \frac{\varepsilon_0 L_{tot}}{I_y} = \frac{P_\varepsilon}{P}, \quad P_\varepsilon = \frac{\varepsilon_0 L_{tot}}{C_m}. \tag{19}$$

The offset-equivalent load P_ε is the vehicle load at which a residual offset ε_0 produces a 100 % bias. It combines the sensor location, the achievable baseline stability and the weight of the lightest vehicle to be screened in one number, and it is the dimensional complement of N_m and R_m.

Table 7 evaluates Eq. (19) for the laboratory configurations. An offset equal to the transducer resolution already biases the load by 0.4–1.0 %, comparable to the total errors of Section 3 and larger than any plausible random noise. Because the bias is inversely proportional to I_y, and therefore to C_m and to the load, it is smallest for heavy vehicles, which suits overload screening, and largest at near-blind locations, which is a second reason to avoid them.

Table 7. Bias of the identified load caused by a zero offset ε_0 in the laboratory, Eq. (19).

Configuration	I_y (mm²)	L_{tot} (mm)	ε_0 = 0.005 mm	0.02 mm	0.05 mm
Single load	2513	5000	+0.99 %	+3.98 %	+9.95 %
Two loads	4168	5250	+0.63 %	+2.52 %	+6.30 %
Three loads	7789	5750	+0.37 %	+1.48 %	+3.69 %

Table 8 evaluates Eq. (19) at the scale of the Cầu Mây girder, with the constants of Section 4.3 for both calibrations and $L_{tot} = 200$ m, the length of the continuous unit. The test truck's wheelbase of 4.95 m would add 2.5 % to L_{tot} and is neglected. At the centre-span gauge a residual offset of 0.05 mm biases a 25.06 t truck, the mean of the weighed test trucks, by 10–11 % under either calibration. At the side-span gauge the bias is 35 % under calibration B and 93 % under calibration A, and SLM does not remove it, because the offset term survives in Eq. (18). The offset values are of LVDT class. Absolute deflection on a long span over water would usually come from a non-contact technique, and the residual offset of that system, after pre/post re-zeroing, has to be established before P_ε can be evaluated.

583 **Table 8.** Offset-equivalent load P_ε, Eq. (19), and bias for the mean weighed test truck, 25.06 t
584 (245.8 kN), at the Cầu Mây gauges; $L_{tot}=200$ m.

Gauge	Calibration	C_m (m²/kN)	P_ε, ε_0 = 0.02 mm (kN)	P_ε, ε_0 = 0.05 mm (kN)	Bias, 0.02 mm	Bias, 0.05 mm
N8, 0.4L	A	4.38×10^{-5}	91.4	228.5	37 %	93 %
N8, 0.4L	B	1.15×10^{-4}	34.9	87.2	14 %	35 %
N7, mid	A	3.61×10^{-4}	11.1	27.7	4.5 %	11 %
N7, mid	B	4.05×10^{-4}	9.9	24.7	4.0 %	10 %

585 ***6.3. Random noise and the SLM reference***

586 A Monte Carlo study with 200 000 realisations per case (Appendix B) gives two results.
587 Random noise is negligible: with σ_a equal to half the resolution, the standard deviation of the
588 identified load is 0.07 %, because integration averages independent perturbations. SLM
589 cancels common-mode error but not independent error. For independent perturbations of the
590 two integrals the relative variances add, and under equal relative standard deviations Eq. (20)
591 holds:

$$\frac{\sigma_{P,SLM}}{P}=\sigma_r\sqrt{1+\frac{1}{N_{rep}}}, \tag{20}$$

592 where N_{rep} is the number of averaged reference crossings. In the present tests the reference
593 integral is only 60 % of the two-load one, and the observed factor is 1.47 rather than the $\sqrt{2}$ of
594 Eq. (20) with $N_{rep}=1$: the standard deviation under the combined model rises from 2.11 %
595 with an averaged reference to 3.09 % with a single one, while BPM stays at 2.12 %.
596 Averaging five or more reference crossings with a reference load comparable to the screened
597 vehicles makes SLM better than BPM in bias and equal in scatter.

598 ***6.4. Traverse speed***

599 Eq. (2) is an integral over space, whereas a monitoring system records time. For a constant
600 speed, $I_y=v\,I_t$ with $I_t=\int y_m\,dt$, so the relative speed error enters the load one-for-one. This
601 term is absent from the static laboratory tests and is likely to be the largest single field
602 contributor. For a single axle on a two-span beam, the influence line vanishes at $x=l$ and
603 $x=2l$, so the interval between these two zero crossings gives $v=l/(t_2-t_1)$. We propose,

without having tested it, to use this interior zero crossing for speed calibration: it is a clean sign change, whereas entry and exit are defined by a signal emerging from the noise floor, and a simply supported span has no interior zero. Using entry and exit instead, the distance is $2l$ and $v = 2l/(t_{out} - t_{in})$; conflating the two distances costs a factor of two in the load. For a convoy the zero crossings shift, and the speed should be obtained from independent axle detection or from a least-squares fit of Eq. (1) with v and the spacings as parameters. Two vehicles on the structure at once return their combined weight, so overlapping crossings must be discarded.

6.5. Quasi-static assumption

The laboratory programme cannot test the quasi-static premise. It was therefore quantified for a constant force crossing the specimen, which isolates bridge dynamics and is not a vehicle–bridge interaction analysis. The beam was modelled with a consistent mass matrix (1.464 kg/m) and Rayleigh damping of 2 % at the first and third modes. The computed fundamental frequency, 11.942 Hz, agrees to four figures with the analytical value. The equations of motion were integrated with the Newmark average-acceleration scheme; refining from 1000 to 16 000 steps changed the error by 0.002 percentage points. Fig. 10 shows the error of the quasi-static integral over 47 speeds spanning $T_{cr}/T_1 = 2lf_1/v$ from 0.6 to 120, and Table 9 its envelope, taken from a denser sweep of 300 speeds so that it does not depend on the speed sampling.

Table 9. Envelope of the dynamic error of the quasi-static estimator.

$T_{cr}/T_1 \geq$	2	3	4	6	8	10	15	20
max ¿ E_{dyn} ∨ ¿ (%)	17.6	8.6	4.1	1.8	0.99	0.59	0.23	0.13

The error oscillates with speed, as the moving-load literature leads one to expect [46], so the envelope is the appropriate summary:

$$\frac{T_{cr}}{T_1} \geq 8 \Rightarrow |E_{dyn}| \lesssim 1\%; \quad \frac{T_{cr}}{T_1} \geq 4 \Rightarrow |E_{dyn}| < 4.2\%. \tag{21}$$

Below $T_{cr}/T_1 \approx 2.5$ the error exceeds 15 % and changes sign, and the estimator should not be used. The dominant mechanism is truncation: at high speed the beam has not returned to its datum when the load leaves (34.5 % of the static peak, with opposite sign, at $T_{cr}/T_1 = 1.5$). Extending the window by four fundamental periods reduces the error by 20–28 %. The error

is not a residue of free vibration that damping would remove: repeating the analysis with damping ratios from 0.5 % to 5 % changes it monotonically. Invariance survives the dynamics approximately: for a fixed total load carried by one axle or by two or three equal axles at spacings of 0.5, 1.0 or 1.5 m, the spread of the integrals is 0.23 % at $T_{cr}/T_1 = 11.9$ and 2.31 % at 3.0. Because the analysis omits the vehicle's own suspension and road roughness, Eq. (21) is a necessary condition for real traffic, not a sufficient one. As an order-of-magnitude orientation only, plausible fundamental frequencies for individual spans of 10 to 30 m give T_{cr}/T_1 of 7 to 20 at 50 to 80 km/h. The ratio is governed mainly by speed and should be checked with the measured f_1 of the structure rather than converted into a speed limit.

6.6. Combined budget

Table 10 assembles an indicative budget for the laboratory tests and for a plausible field deployment on a short-span concrete continuous bridge with a well-conditioned gauge. Following the GUM [47], known biases are corrected rather than budgeted: the trapezoidal bias is removed by Simpson quadrature or by fine sampling (Section 2.4). All remaining contributions are treated as independent standard uncertainties and combined by root-sum-square, with the same rule in every column. The field entries are engineering scenarios propagated through the sensitivities derived above, with environmental drift values from long-term monitoring [48,49]. They are not a metrological validation of a complete system. The Type A entries are upper bounds set by the transducer resolution and the Monte Carlo result of Section 6.3, not evaluations from the repeated cycles.

Table 10. Indicative sensitivity-based uncertainty budget (standard uncertainties in per cent, combined by root-sum-square in every column; a bound "< x" is entered as x, and the dynamic term as 0 at the lower and 1 % at the upper end of a range).

Contribution	Type	Sensitivity	Laboratory (SLM)	Field (SLM)	Field (BPM)
Flexural rigidity EI	B	1	cancels	cancels	5–15 %
Span length l	B	4	cancels	cancels	0.4–2 %
Sensor gain γ	B	1	cancels	cancels	1–2 %
Quadrature bias e_q	B	1	corrected	corrected	corrected
Reference weighing P_0	B	1	0.1 %	0.5–1 %	n/a
Residual zero offset ε_0	B	L_{tot}/I_y	0.4–1 %	2–10 %	2–10 %
Traverse speed v	B	1	n/a	1–3 %	1–3 %
Dynamic effects ($T_{cr}/T_1 \geq 8$)	B	1	n/a	< 1 %	< 1 %

Contribution	Type	Sensitivity	Laboratory (SLM)	Field (SLM)	Field (BPM)
Random noise σ_a	A (bound)	< 0.1	< 0.1 %	0.1–0.3 %	0.1–0.3 %
Reference scatter (N_{rep} = 1)	A (bound)	1	< 0.1 %	0.1–0.3 %	n/a
Combined (root-sum-square)			**0.4–1.0 %**	**2.3–10.5 %**	**5.6–18.5 %**

In the laboratory, SLM is limited by zero offset and reference weighing, and the combined 0.4–1.0 % is consistent with the observed deviations of 0.61 % and 0.70 %. In the field, baseline stability and speed dominate. The field offset entry of 2–10 % assumes a well-conditioned gauge; at an ill-conditioned location the offset term alone can be several times larger, as Table 8 shows for the side-span gauge of a long continuous girder. The ranges are scenario-based and must be verified for each structure and instrumentation chain. They fall short of a certified weighing device, and the method is proposed for preliminary overload screening, with the decision threshold set sufficiently apart from the legal limit and the design live loads [50,51]; establishing that threshold requires field calibration and lies outside this study.

7. Recommendations and limitations

7.1. Recommendations for sensor placement and operation

The results above combine into a pre-installation procedure.

1. Compute C_m for each candidate location: from Eq. (4) for two spans, Eq. (8) for symmetric three spans, or Eq. (3) with one static analysis of a model with $EI(x)$, or one static uniform-load test.

2. Check the blind locus. For prismatic beams, stay out of the band $|N_m| < 0.2$ around the blind locus, on both sides of it: for two spans with uniform stiffness and a mid-span sensor this excludes $1.385 < \rho_L < 1.525$, and for symmetric three spans with a mid side-span sensor $1.621399 < \beta < 1.775$. A negative constant outside the band is observable; it needs sign-reversed processing and depends mainly on the adjacent span.

3. For haunched girders, report N_m with its EI_{ref} and also $R_m = C_m / A^{+¿¿}$, and rank locations by $|R_m|$. Evaluate C_m over a plausible range of pier stiffness; the pinned and fixed limits only bound it. A location whose C_m changes strongly within that range, such as the side-span gauge of Section 4, needs an in-situ calibration.

4. Place the sensor near ξ^{i} (0.42154 of the span for two equal spans; Table 2 for three spans) or, on long-main-span girders, in the main span.

5. Use SLM with a weighed vehicle comparable to the screened traffic, averaging at least five reference crossings. Use BPM only where the location is well conditioned and EI and the support conditions are known. For side-span gauges on continuous girders, take C_m from SLM or from a model identified in situ, never from the design model.

6. Evaluate $P_\varepsilon = \varepsilon_0 L_{tot}/C_m$, Eq. (19), for the residual offset that the instrumentation can achieve after re-zeroing, and compare it with the lightest vehicle to be screened.

7. Check $T_{cr}/T_1 = 2lf_1/v \geq 8$ for the actual frequency, not a speed limit.

8. Integrate over a window tight around the crossing, re-zero the baseline before and after each crossing, and reject records whose end values differ by more than a set tolerance. Remove dynamics with a low-pass filter well below f_1, never with a high-pass filter.

9. Use Simpson quadrature, with a 3/8 end panel for an odd number of intervals, only when the entry and exit of every axle fall on panel edges; otherwise use the trapezoidal rule with a sampling interval of $l/20$ or finer, and obtain the speed from independent axle detection or from a fitted response model.

The procedure positions the deflection integral as a reduced alternative to BWIM rather than a competitor. It returns only the gross weight of a load group, needs one displacement channel and only the scalar C_m instead of an influence line or a full model, and involves a single division rather than a regularised inversion. Its natural role is low-cost screening across many secondary structures, with instrumented BWIM or enforcement weighing reserved for the vehicles it flags, and with the screened loads feeding the load models used in assessment. With several sensors, each crossing gives a vector of integrals proportional to the same load, a one-parameter model fitted by weighted least squares in which a near-blind sensor carries almost no weight; the maps of Figs. 2 and 3 then double as sensor-selection charts.

7.2. Limitations

The laboratory tests used one small steel specimen with idealised supports and statically positioned loads; concrete cracking, creep, bearing friction and moving traffic are not represented. The accuracy class of the calibrated weights is not reported [NEED VALUE],

and the Type A component was bounded rather than evaluated from the repeated cycles. The invariance is verified only indirectly, because the configurations differ in both load and spacing, and the measured-to-analytical ratio rises from 1.010 to 1.017 with the load group. A direct test with a fixed total load at several spacings and load magnitudes needs no new apparatus, and the predicted blind configuration (spans of 2500 mm and 3644 mm on the same rig) is a falsifiable prediction; neither has yet been carried out. The dynamic study models a constant force and omits vehicle–bridge interaction and road roughness. The treatment is one-dimensional, so transverse load position is ignored; the eccentric readings at Cầu Mây show that it matters.

The bridge application rests on static parked-convoy data, the mean mass of the weighed trucks rather than the masses of the trucks in each convoy, and an assumed depth law, and two static readings cannot separate the girder rigidity from the pier stiffness; the designer's FE model used for calibration A is itself a model. The side-span result is therefore stated as a range, not as a single value. A second three-span box girder (42 + 63 + 42 m) was examined, but its two-dimensional model could not reproduce the deflections reported for the designer's FE model (misfits of −11 %, +22 % and −12 %), and cross-section and pier dimensions are missing; it is therefore not analysed here. Repeating the analysis with the full design models, pier drawings or measured natural frequencies, and testing Eq. (2) with crawl passes of weighed vehicles on a continuous girder, are the next steps.

8. Conclusions

This paper examined when the total load on a continuous girder is observable from the integral of a single deflection channel. The integral over a complete crossing equals the total load times a constant C_m of the structure and the sensor position, a property that underlies influence-area weigh-in-motion. By reciprocity, C_m is the deflection at the sensor under a uniform unit load, which makes it available before any instrument is installed.

This characterisation gives closed forms for continuous beams. For a two-span beam of uniform stiffness the blind span ratio is $(3+\sqrt{33})/6 = 1.45743$ and the optimal sensor position is $(1+\sqrt{33})/16 = 0.42154$ of the span. For a symmetric three-span beam instrumented at mid side span, the blind main-to-side ratio is $\beta = 1.700594$, the near-blind band runs from 1.621399 to 1.775, and $2\cos(\pi/9) = 1.879385$ ends the side-span blind locus: beyond it every side-span position has a negative but non-zero constant, dominated by the main span.

On a laboratory two-span beam with statically positioned loads, two- and three-load groups were identified within 1.19 % by BPM and 0.70 % by SLM against direct weighing. The entire difference between the two methods was the calibration error of the reference record.

On the 55 + 90 + 55 m Cầu Mây box girder, the observability of the side-span gauge could not be identified from design data and two static readings. Two calibrations reproduce the measured deflections and give indicators of 0.18 and 0.48, with 82 % and 44 % of the positive influence area cancelled. The centre-span gauge is stable at 0.21–0.23. A residual offset of 0.05 mm would bias a 25.06 t truck by 35–93 % at the side-span gauge and by 10–11 % at the centre-span gauge. Side-span gauges on continuous box girders are ill-conditioned with respect to pier rotational stiffness, so their constant must come from a weighed-vehicle calibration or an in-situ identified model, never from the design model. This application uses static data and is not a validation of the integral on a continuous bridge.

The error budget places the residual zero offset, not random noise, at the top of the field error hierarchy, and crossing records from simply supported spans illustrate that baseline handling can change the integral by tens of per cent while low-pass filtering leaves it intact. For a constant moving force the quasi-static error is about 1 % at a crossing time of eight fundamental periods. Computing $|C_m|$, $C_m/A^{+¿¿}$ and P_ε for each candidate location is proposed as a routine step before installing a deflection sensor for load screening on continuous bridges.

Appendix A. Two-span beam: influence line and closed form

For two equal spans l with a knife-edge interior support, taking the support moment as redundant gives $M_B = -P\,x\left(l^2 - x^2\right)/\left(4\,l^2\right)$ for a load on the first span. The reactions for a load at $0 \le x \le l$ are

$$R_A = P\,\frac{x^3 - 5\,l^2\,x + 4\,l^3}{4\,l^3},\; R_B = P\,\frac{-x^3 + 3\,l^2\,x}{2\,l^3},\; R_C = P\,\frac{x^3 - l^2\,x}{4\,l^3}, \tag{A.1}$$

and for $x = l/2$ Eq. (A.1) gives the classical values $13\,P/32$, $11\,P/16$ and $-3\,P/32$. Double integration on $[0, x)$, $[x, l)$ and $[l, 2l)$ with six displacement conditions ($y(0) = 0$; continuity of y and y' at the load; $y(l) = 0$ from each side; $y(2l) = 0$) gives the influence line at $s = l/2$:

$$\eta(x)=\frac{1}{EI}\begin{cases}\dfrac{x\left(9l^2-13x^2\right)}{192}, & 0\le x\le l/2,\\[2mm] \dfrac{(x-l)\left(4l^2-29lx+19x^2\right)}{192}, & l/2<x\le l,\\[2mm] \dfrac{(2l-x)\left(3l^2-4lx+x^2\right)}{64}, & l<x\le 2l.\end{cases} \tag{A.2}$$

The influence line neglects the 400 mm of overhang beyond the end supports. Eq. (A.2) is continuous and twice differentiable at $x=l/2$ and $x=l$; the third branch carries the denominator $64\,EI$, not $192\,EI$.

At mid-span the deflection under a mid-span load is $2.875\,Pl^3/(192\,EI)$, 71.9 % of the simply supported value; the maximum over load position, at $x=0.480\,l$, is 72.1 %.

For unequal spans, the uniform-load support moment is $M_B=-q_0\left(L_1^3/EI_1+L_2^3/EI_2\right)/\left[8\left(L_1/EI_1+L_2/EI_2\right)\right]$. Superposing the simply supported uniform-load deflection of span 1 and the deflection due to M_B, evaluating at x_m and dividing by q_0 gives Eq. (4) with $\mu=-8\,M_B/\left(q_0L_1^2\right)$, which is Eq. (5). Differentiating μ with respect to ρ_L gives a numerator $\rho_{EI}\left(3\rho_L^2-1\right)+2\rho_L^3$; for $\rho_{EI}=1$, μ falls to 0.750 at $\rho_L=1/2$, passes 1 at equal spans and reaches 3 at $\rho_L=2$. Table A.1 lists the normalised constant against the span ratio for a mid-span sensor.

The blind condition has a mechanical reading. In units of $L_1^4/(192\,EI_1)$ and for a mid-span sensor, the positive contribution of the instrumented span, $C_m^{+¿¿}$, varies only between 1.50 and 2.00 for ρ_L from 0.5 to 2.0, whereas the negative contribution of the adjacent span, C_m^-, sweeps from −0.125 to −4.00; the degeneracy is driven almost entirely by the adjacent span. For equal spans $C_m^{+¿=1.750¿}$ and $C_m^-=-0.750$, so the net constant is only 4/7 of the positive part. Moving the sensor from $0.50\,L_1$ to $0.42\,L_1$ lowers the positive contribution only from 1.750 to 1.732 but the magnitude of the negative one from 0.750 to 0.692, so the net constant rises from 1.000 to 1.040. The optimum therefore lies towards the outer support, where the adjacent span interferes less; on a simply supported beam mid-span is optimal by symmetry, so the shift is a signature of continuity.

Table A.1. $192\,EI_1C_m/L_1^4$ against the span ratio, sensor at mid-span, uniform stiffness.

L_2/L_1	0.5	0.7	0.8	1.0	1.2	1.3	1.4	1.45743	1.6
Value	1.375	1.315	1.240	1.000	0.640	0.415	0.160	0	−0.440

Appendix B. Monte Carlo robustness analysis

The study used 200 000 realisations of Eq. (16) per case: gain error alone (σ_γ = 2 %), zero offset alone (σ_ε = 0.005 mm), random noise alone (σ_a = 0.0025 mm) and all three combined. Each realisation perturbed the measured record, recomputed the integral with the same rule and grid and re-identified the load. For SLM, an averaged (noise-free) reference and a single noisy reference were distinguished. Table B.1 gives the two-load results. For the three-load case the bias is 0.08–0.09 percentage points higher and the standard deviations agree within 0.07 percentage points, except under zero offset alone, where the smaller offset sensitivity L/I_y lowers the standard deviation from 0.64 % to 0.37 % (BPM). The bias column reflects the deviation of the measured record from weighing (Table 4) and is constant across error sources; the information is in the spread.

Table B.1. Monte Carlo results, two-load configuration (relative error with respect to weighing, %).

Error source	Method	Reference	Bias	Std. dev.	p5	p95
Gain, σ_γ = 2 %	BPM	n/a	+1.10	2.01	−2.21	+4.40
	SLM	averaged	+0.61	2.00	−2.69	+3.90
	SLM	single	+0.66	2.85	−3.94	+5.41
Zero offset, σ_ε = 0.005 mm	BPM	n/a	+1.10	0.64	+0.05	+2.15
	SLM	averaged	+0.61	0.63	−0.43	+1.65
	SLM	single	+0.62	1.19	−1.31	+2.60
Random, σ_a = 0.0025 mm	BPM	n/a	+1.10	0.07	+0.99	+1.21
	SLM	averaged	+0.61	0.07	+0.50	+0.72
	SLM	single	+0.61	0.13	+0.40	+0.82
Combined	BPM	n/a	+1.10	2.12	−2.39	+4.59
	SLM	averaged	+0.61	2.11	−2.87	+4.08
	SLM	single	+0.65	3.09	−4.34	+5.83

Appendix C. Laboratory data

Table C.1. Mid-span deflection of span AB (mm), mean of three cycles. Single load 40.64 N; analytical values from Eq. (A.2).

x (mm)	Analytical, single	Measured, single	Two loads	Three loads
0	0.0000	0.0000	0	0

x (mm)	Analytical, single	Measured, single	Two loads	Three loads
250	0.8874	0.8844	0.4625	0.5525
500	1.6968	1.6975	1.8813	1.0450
750	2.3501	2.3525	3.1413	2.5413
1000	2.7693	2.7738	4.0988	4.8700
1250	2.8764	2.8827	4.6350	6.7588
1500	2.6253	2.6363	4.6238	7.9350
1750	2.0980	2.1100	4.0688	8.2775
2000	1.4087	1.4213	3.1375	7.6825
2250	0.6713	0.6838	1.9600	6.2800
2500	0.0000	0.0000	0.7715	4.3725
2750	−0.5132	−0.5069	−0.2689	2.2975
3000	−0.8644	−0.8556	−1.0063	0.3021
3250	−1.0715	−1.0600	−1.4988	−1.2588
3500	−1.1526	−1.1438	−1.7813	−2.3650
3750	−1.1255	−1.1200	−1.8600	−3.0363
4000	−1.0085	−1.0013	−1.7738	−3.3300
4250	−0.8194	−0.8175	−1.5525	−3.2663
4500	−0.5763	−0.5816	−1.2213	−2.9450
4750	−0.2971	−0.3019	−0.8125	−2.4125
5000	0.0000	0.0000	−0.3325	−1.7088
5250			0	−1.0713
5500				−0.3650
5750				0

Appendix D. Three-span beams: main-span sensor, verification and asymmetric layouts

For a sensor at $\eta = x/L_m$ in the main span of the symmetric beam of Section 2.3 the constant is $192\,E\,I_m\,C_m/L_m^4 = 8\left[\eta\left(1-2\eta^2+\eta^3\right)-\nu\,\eta\left(1-\eta\right)\right]$, with $\nu = 3\left(\rho_{EI}+\beta^3\right)/\left[\beta^2\left(2\,\rho_{EI}+3\,\beta\right)\right]$. The symmetric-beam expressions of Section 2.3 agree with FE evaluations of Eq. (3), by the uniform-load route and by integration of the influence line, to a maximum relative error of 7.5 × 10^{-12} over 60 comparisons (six (β, ρ_{EI}) pairs, five sensor positions, both routes). The cubic for the optimal side-span position in Section 2.3 reduces for two spans to the quadratic of Eq. (7), because $\mu = 1$ makes $\xi = 1$ a root; for three spans it does not.

For arbitrary L_1, L_2, L_3 and $E\,I_1, E\,I_2, E\,I_3$, the two support moments follow from the three-moment equations at B and C, solved symbolically; the sensor deflection under a uniform load then follows by superposition, as in Appendix A. The general expressions are long and are provided in the supplementary scripts. Four asymmetric cases agree with FE to a relative error of at most 9.2 × 10^{-12}. With the sensor at mid span 1 and uniform stiffness, the blind

ratio L_1/L_2 is 0.589774, 0.591969, 0.587260, 0.565712, 0.525757 and 0.461410 for L_3/L_2 = 0.3, 0.5, 0.6, 0.8, 1.0 and 1.2.

Computational Implementation

All analyses were implemented by the authors in reproducible scripts: the symbolic verification of the two-span solution, the closed forms of Sections 2.2 and 2.3, the quadrature and Monte Carlo studies, the bridge model of Section 4, the record processing of Section 5 and the moving-load analysis of Section 6.5. Generative artificial intelligence was used as an assistance tool during code development and debugging. Every equation and numerical output was checked independently against an analytical, finite element or convergence-based route, and the scripts and the console output of each run are provided as supplementary material.

Data Availability

The laboratory deflection data supporting the findings are given in full in Appendix C. The derived data behind every figure and table, the analysis scripts and the console output of each run are provided as supplementary material. The load tests that produced the Cầu Mây acceptance-test data of Section 4 and the field crossing records of Section 5 were performed by an author of this paper, and the authors are permitted to publish the processed data. The processed data (static deflection readings, truck positions and weighed truck masses for Cầu Mây, and the processed crossing records of Section 5) are available at [NEED VALUE: repository name and DOI].

Conflicts of Interest

The Cầu Mây load test analysed in Section 4 was performed by the authors' team as part of inspection work carried out by Công ty TNHH Tư vấn Đại học Xây dựng (HUCE consultancy) [NEED VALUE: name(s) of authors involved and nature of relationship]. The authors declare that there is no other conflict of interest regarding the publication of this article.

Funding Statement

This research did not receive any specific grant from funding agencies in the public, commercial or not-for-profit sectors. [NEED VALUE: name the employer under whose employment the work was performed, and the source of the article processing charge, if any.]

Author Contributions

Quoc-Bao Nguyen: Conceptualization, Methodology, Formal analysis, Software, Writing – original draft. **Thai-Son Vu:** Conceptualization, Investigation, Validation, Supervision, Project administration, Writing – review & editing. **Hung-Son Nguyen:** Investigation, Data curation, Visualization, Writing – review & editing.

Disclosure of Generative AI Use

During the preparation of this work the authors used Claude (Anthropic) to assist with code development and debugging and with language editing. All scientific formulations, procedures, numerical results and interpretations were independently reviewed and verified by the authors, who take full responsibility for the content of the manuscript.

Acknowledgments

The authors gratefully acknowledge the Laboratory of Transportation Technology and Systems, Faculty of Bridge and Road Engineering, Hanoi University of Civil Engineering, for providing the test facilities used in this study. [NEED VALUE: whether to acknowledge Hai Duong PMU and the testing consultant for access to the acceptance-test report.]

Supplementary Materials

The supplementary material contains: (i) the analysis scripts and the console output of each run, covering the symbolic verification of the closed forms, the general asymmetric three-span expressions of Appendix D, the quadrature and Monte Carlo studies, the Cầu Mây model and calibrations of Section 4, the record processing of Section 5 and the moving-load analysis of Section 6.5; (ii) the derived data behind every figure and table; (iii) the fitted speeds and speed-normalised integrals of the field crossing records of Section 5; (iv) Figure S1, the indicators N_m and R_m of the Cầu Mây gauges for calibrations A and B with the pinned … fixed bounds (the values of Table 5 in graphical form).

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1009 **Figures**

1010 *[Figure 1 to be inserted: source file supplied by the authors]*

1011 **Fig. 1.** Continuous beams considered: (a) two-span beam with instrumented span L_1, adjacent
1012 span L_2 and sensor at ξL_1; (b) symmetric three-span beam with side spans L_s, main span
1013 $L_m = \beta L_s$ and side-span sensor at ξL_s.

1014 *[Figure 2 to be inserted: source file supplied by the authors]*

1015 **Fig. 2.** Load observability of a two-span beam: (a) normalised constant $192\,E I_1 C_m / L_1^4$ over
1016 the span and stiffness ratios for a mid-span sensor, with the closed-form blind locus of Eq. (6)
1017 (thick line) and the near-blind band $\left|192\,E I_1 C_m / L_1^4\right| < 0.2$ (hatched); (b) dependence on sensor
1018 position for several span ratios, with the optimum $\xi^{*} = \left(1+\sqrt{33}\right)/16$ for the symmetric case.

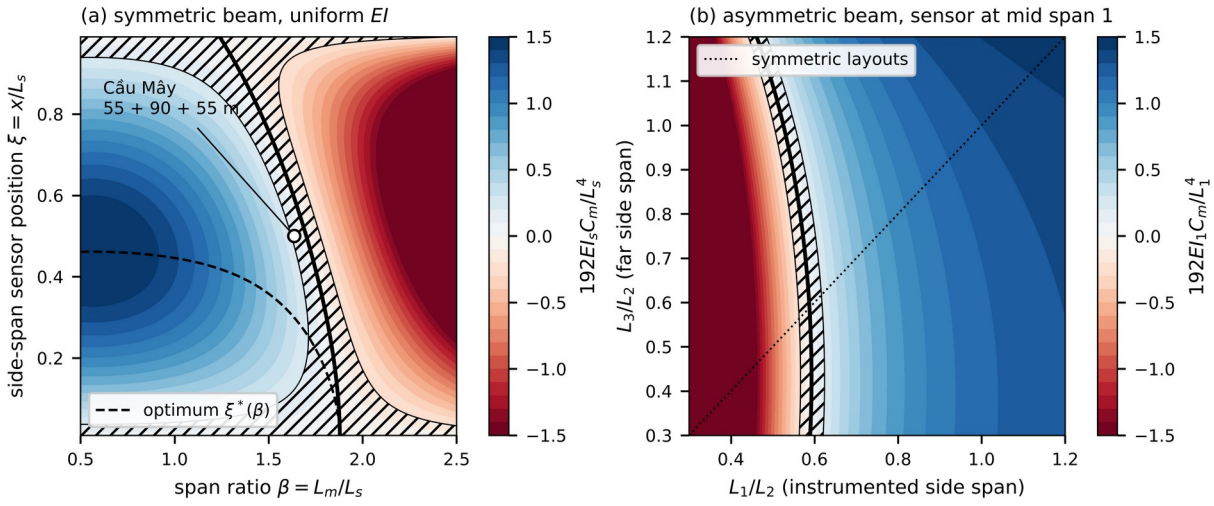


Fig. 3. Load observability of a three-span beam with uniform rigidity: (a) symmetric beam, $192\,E\,I_s\,C_m/L_s^4$ over β and the side-span sensor position ξ, with the blind locus (thick line), the near-blind band (hatched), the optimum $\xi^*(\beta)$ (dashed) and the Cầu Mây layout with a mid side-span sensor (open circle); (b) asymmetric beam, sensor at mid span 1, over L_1/L_2 and L_3/L_2, with the symmetric layouts dotted. Colours are clipped at ±1.5.

[Figure 4 to be inserted: source file supplied by the authors]

Fig. 4. Laboratory specimen: (a) elevation and instrumentation; (b) general view of the test rig.

[Figure 5 to be inserted: source file supplied by the authors]

Fig. 5. Mid-span deflection of span AB against the position of a single 40.64 N load: analytical, FE and measured. The lower panel gives the absolute difference between measurement and theory. Shaded areas are the positive and negative contributions to the deflection integral.

[Figure 6 to be inserted: source file supplied by the authors]

Fig. 6. Measured deflection integral against total applied load for the three configurations, with the theoretical slope $l^4/(192\,E\,I)$.

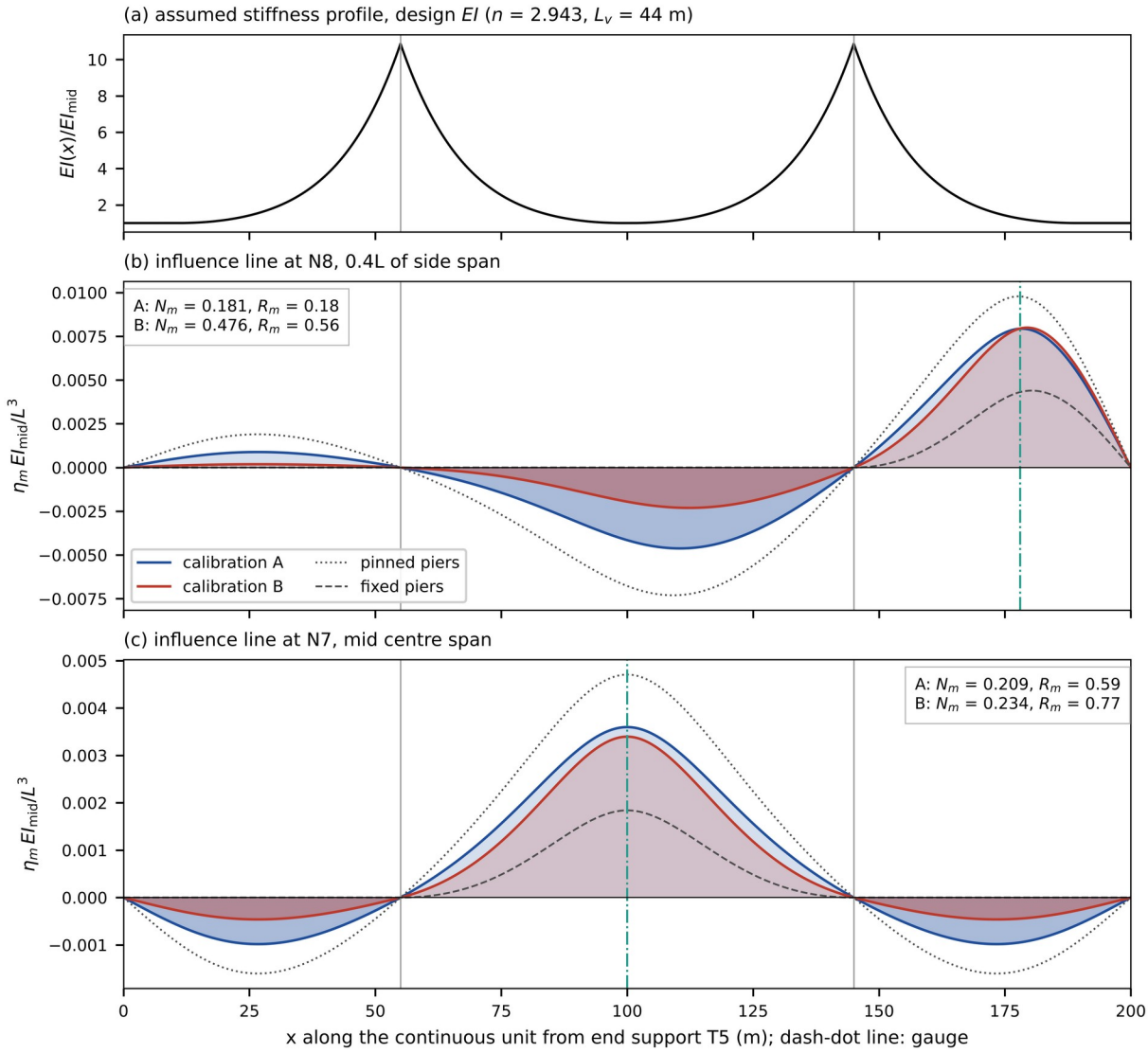


Fig. 7. Cầu Mây main bridge (55 + 90 + 55 m): (a) assumed stiffness profile $EI(x)/EI_{mid}$ ($n=2.943$, $L_v=44$ m); (b), (c) deflection influence lines, normalised by EI_{mid}/L^3, at the installed gauges N8 (0.4L of the side span) and N7 (mid centre span) for calibration A (design EI, k_{FE}), calibration B ($\alpha=0.716$, $k=1.2\times10^8$ kN m/rad), pinned piers and fixed piers (design EI). Positive and negative areas are shaded for calibrations A and B; the boxes give N_m and R_m of Table 5; the dash-dot line marks the gauge.

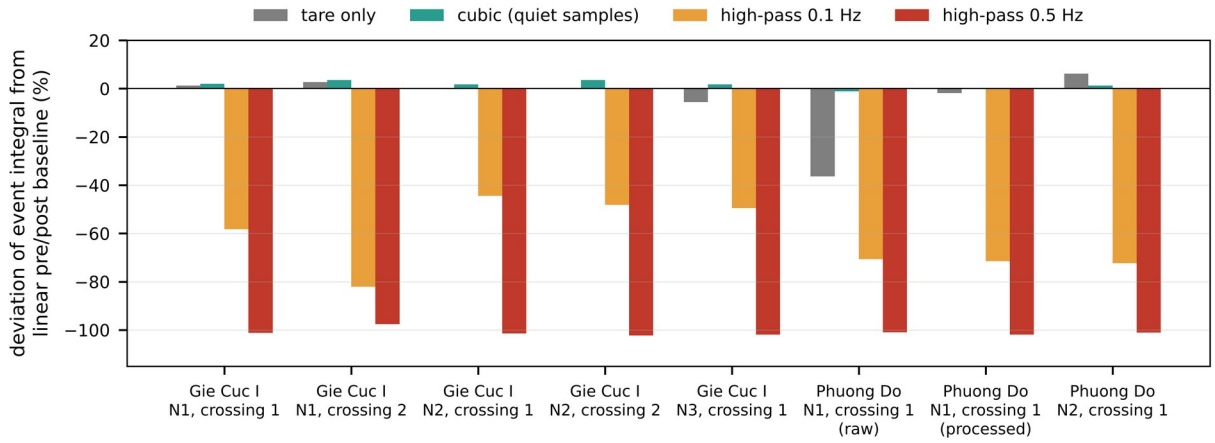


Fig. 8. Deviation of the event deflection integral from the linear pre/post baseline for the seven complete crossings (raw channel; both channels for Phuong Do N1), for tare-only, cubic and high-pass (0.1 Hz and 0.5 Hz) baselines.

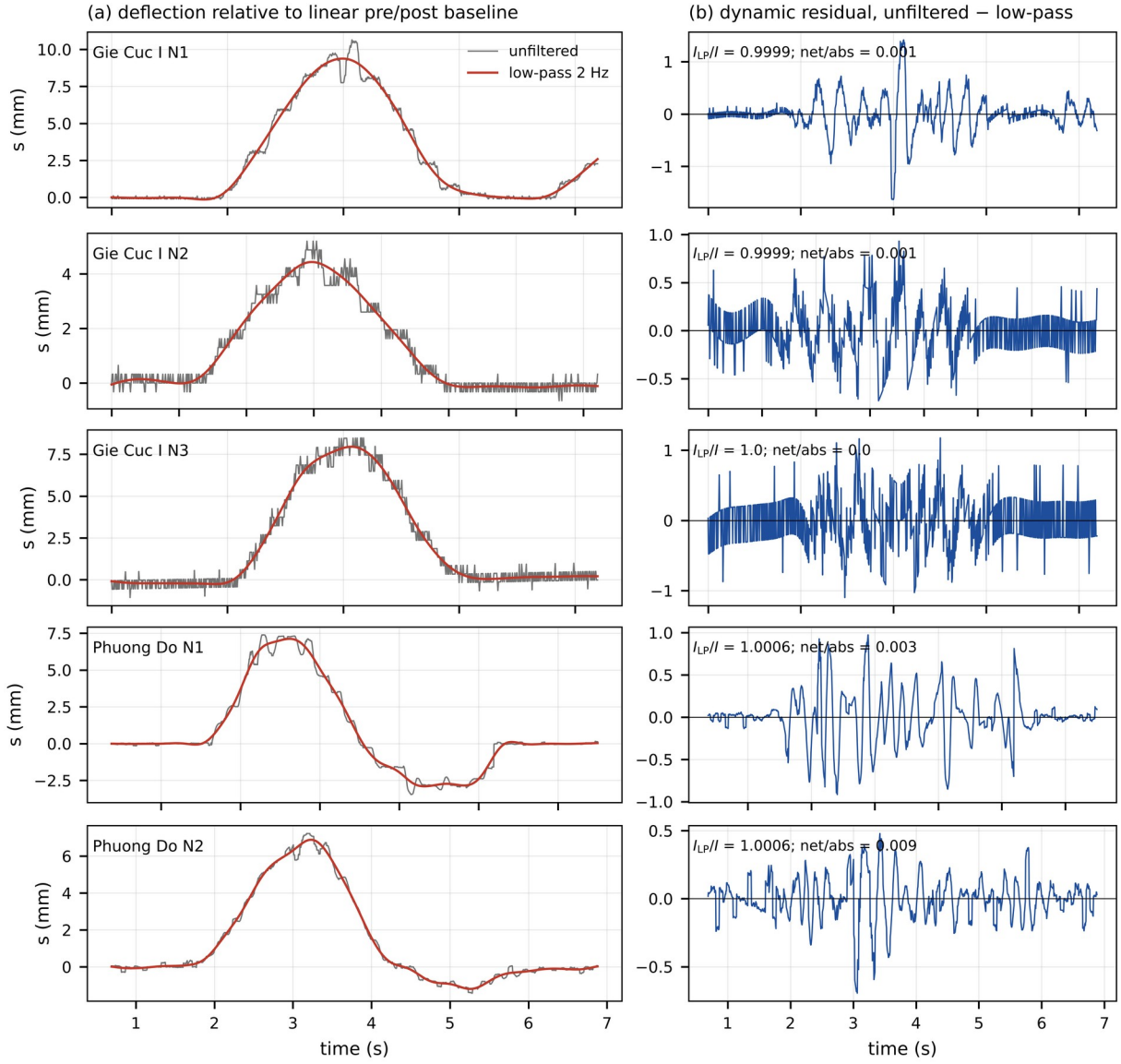


Fig. 9. First complete crossing on each span (processed channel): (a) deflection relative to the linear pre/post baseline before and after a 2 Hz zero-phase low-pass filter; (b) dynamic residual, with the ratio of filtered to unfiltered event integrals and the ratio of the net to the absolute integral of the residual.

[Figure 10 to be inserted: source file supplied by the authors]

Fig. 10. Error of the quasi-static deflection-integral estimator for a constant force crossing the laboratory beam, against the ratio of crossing time to fundamental period. Shaded bands mark ±1 % and ±5 %.